ANSWER KEY

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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Multi-Disciplinary Courses (MDC) - MG1MDCPOL101 - ECOLOGICAL GOVERNANCE

(2024 ADMISSION ONWARDS)

PART A

1. b). 17

2. c). Reforestation

3. c). Focus on long term targets

4. b). From the urban canyon effect trapping long-wave radiation

5. b). Through validating emission reduction goals aligned with Paris Agreement limits

6. a). 2015

7. c). United Nations Framework Convention on Climate Change (UNFCCC) Secretariat

8. b). Paris Agreement

9. b). 30x30 Initiative

10. b). To halt and reverse biodiversity loss by 2030

PART B

11. Marginalized communities face higher exposure to industrial pollution- waste dumping- unsafe drinking water- inadequate sanitation- often live near landfills- highways, and hazardous industries- resulting in greater health risks- limited green spaces- unequal access to environmental resources and disaster protection.

12. UNFCCC was established in 1992-coordinate global efforts against climate change- provides a platform for negotiation-including the Kyoto Protocol and Paris Agreement- facilitates climate finance and adaptation support- promotes emission monitoring- encourages cooperation to limit global temperature rise and strengthen climate resilience.

13. Common ozone-depleting substances- include chlorofluorocarbons, halons- carbon tetrachloride-methyl chloroform-these chemicals release chlorine and bromine in the stratosphere- breaking down ozone molecules- reduction under the Montreal Protocol has helped restore the ozone layer and prevent harmful ultraviolet radiation impacts.

14. Wetland drainage exposes organic-rich soils to oxygen- increasing microbial decomposition and releasing methane- converting wetlands from methane sinks to sources-

reduces water saturation- alters vegetation, disrupts carbon storage- accelerates greenhouse gas emissions-undermining wetlands' natural climate-regulating functions.

15. NAPCC-guides India's climate response through eight national missions -focusing on solar energy- energy efficiency-sustainable agriculture- water conservation- integrates climate goals with development-promotes low-carbon planning- encourages renewable energy expansion- frames long-term mitigation and adaptation strategies across sectors.

16. MoEF was established in 1985- amid rising environmental concerns- including industrial pollution- deforestation- global environmental debates-influenced by the 1972 Stockholm Conference- growing public awareness-India created a dedicated ministry to regulate environmental protection-wildlife conservation- sustainable development.

17. Kunming-Montreal Framework- supports climate mitigation- by conserving ecosystems that store carbon- restoring degraded landscapes-reducing deforestation- promoting sustainable land use-protecting biodiversity and expanding conservation areas- strengthens natural carbon sinks and enhances ecosystem resilience against climate impacts.

18. Climate change accelerates the spread of invasive species -by altering temperature-rainfall, habitat conditions that favour their expansion-invasives outcompete native species- disrupt ecosystems-reduce biodiversity- intensify ecological stress-posing new challenges for conservation and ecosystem management.

PART C

19. Primary drivers of biodiversity loss- include habitat destruction- climate change-pollution-overexploitation- invasive species- these pressures disrupt food webs- degrade ecosystem functions- reduce species populations- undermine services like pollination- water purification- climate regulation-ultimately threatening human health, livelihoods, long-term ecological stability.

20. Greenhouse effect is the natural process where gases like CO_2 trap heat in the atmosphere-warming the Earth-human activities intensify this effect- leading to climate change- Paris Agreement aims to reduce global emissions- limit temperature rise to well below 2°C-strengthen adaptation- enhance climate finance- promote global cooperation to mitigate its impacts.

21. NAPCC aims to promote sustainable development- through missions on solar energy-energy efficiency- water conservation- sustainable agriculture- seeks low-carbon growth and climate resilience- facilitated renewable energy expansion and awareness- though uneven implementation- limited funding- institutional challenges restrict its overall effectiveness.

22. Biological Diversity Act, 2002- safeguards India's biodiversity- by regulating access to biological resources- protecting traditional knowledge- promoting equitable benefit-sharing-empowers local biodiversity committees- supports conservation strategies- prevents biopiracy-Act strengthens ecosystem protection and recognizes community roles in maintaining ecological balance.